



Toward biophysical synergy: Investigating advection along the Polar Front to identify factors influencing Alaska sablefish recruitment



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ABSTRACT

In fisheries stock assessment, reliable estimation of year-class strength is often hindered by lack of data on early life history stages and limited knowledge of the underlying environmental processes influencing survival through these stages. One solution to improving these estimates of year-class strength or recruitment is to first develop regional indices representing the spatial and temporal extent of a hypothesized feature influencing a species' recruitment. These covariates should then be integrated within a population model where a variety of model selection techniques may be conducted to test for a reduction in recruitment uncertainty. The best selected model(s) may provide insight for developing hypotheses of mechanisms influencing recruitment. Here we consider the influence of a large-scale oceanographic feature, the North Pacific Polar Front, on recruitment of Alaska sablefish (*Anoplopoma fimbria*). Our working hypothesis is that advection of oceanic properties along the Polar Front and associated currents plays a key role in shaping the oceanographic climate of Alaskan waters and, hence, the environment that sablefish encounter during their early life history. As a first step in this investigation, we developed time series of sea surface temperature along the Polar Front mean path. We then integrated this data into the recruitment equations of the sablefish assessment base model. Model selection was based on a multistage hypothesis testing procedure combined with cross-validation and a retrospective analysis of prediction error. The impact of the best model was expressed in terms of increased precision of recruitment estimates and proportional changes in female spawning biomass for both current estimates and in future projections. The best model suggested that colder than average wintertime sea surface temperatures in the central North Pacific represent oceanic conditions that create positive recruitment events for sablefish. The incorporation of this index in the sablefish model provided moderate reduction in unexplained recruitment variability and increased future projections of spawning biomass in the medium term. Based on this result, we developed a conceptual model of three mechanisms that in combination form an ocean domain dynamic synergy (ODDS) which influences sablefish survival through the pelagic early life history stage. Successfully incorporating environmental time series into the sablefish assessment could establish a foundation for future ecosystem-based management and allow for more informed and efficient resource allocation to stakeholders.

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1. Introduction

Year-class strength is a fundamental driver of fluctuations in stock size for many fish populations and is generally considered to be the abundance of the youngest fish entering a population that can be estimated successfully (Myers, 1998; Maunder and Watters, 2003). Developing reliable estimates of year-class strength, which is also termed recruitment, has been a long-

standing problem for fisheries management. The difficulty arises from limited survey or fishery information on the early life stages of the fish. Often direct estimation of early life stages from egg and larval surveys is very limited and standard adult surveys cannot provide estimates of year-class strength until the fish are several years old because the survey does not sample the early juvenile sizes. Therefore, the primary data on these stages are derived from modeling demographic information such as ages and lengths to provide estimates of past recruitment. The uncertainty of these estimates increases toward the beginning and end of a population time series when the demographic data become very limited due to estimation error at advanced ages or no

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information as yet on recently produced year classes (Hanselman et al., 2012). The validity and acceptance of recommendations for future harvests are based on the quality of recruitment estimates. The high uncertainty resulting from the lack of data supporting these estimates usually results in their removal for determining projections of future harvests. If these estimates are actually larger or smaller than average, then catch quotas may be set too low or too high, resulting in unattained yield due to natural mortality or unsustainably high fishing pressure that must inevitably be lowered in the future. This produces an inefficient quota-setting system with unnecessary fluctuations in future catch quotas. Managers run the risk of under- or over-harvesting the species as they react to the inertia of poor information. This is not consistent with optimizing yield (Magnuson-Stevens Fishery Conservation and Management Act, amended 1996).

One potential option to decrease uncertainty in this system is to integrate ecologically plausible environmental time series into a population model (Deriso et al., 2008). Often relationships between recruitment and the environment are evaluated using correlative analyses outside an estimation procedure. Upon re-evaluation with new observations, these relationships do not continue to hold and managers are reluctant to incorporate such environmental time series into harvest recommendations (Myers, 1998). An alternative to simple correlations is to incorporate environmental time series into a statistical catch-at-age analysis where model parameters are estimated simultaneously in a maximum likelihood framework (Maunder and Watters, 2003). In this way, the uncertainty in the environment–recruit relationship and interactions with other data sources may be estimated and conveyed to management. Additionally, environmental influences on species at the center of their geographic distribution may not be as influential or identifiable as for species at their geographic limit (Myers, 1998). The slower growth rates of cooler water species usually extends the duration of life history stages resulting in a potentially greater influence of variability within the physical and biological environment (Field and Ralston, 2005). Environment–recruit relationships developed for stocks at the more extreme edge of their distribution may be more consistent and reliable over time. The caveat of these relationships are the difficulty in identifying the causal forcing factors that influence a given species' survival and then developing proxy environmental time series that represent the appropriate spatial and temporal extent of these factors.

In this study we sample a large-scale oceanographic feature in order to develop environmental indices that represent the spatial and temporal patterns of that feature. These indices are then tested within a population model to determine the approximate location and timing for influencing a given species' recruitment. Improvement in subsequent recruitment estimation may provide insight into developing hypotheses for potential biological mechanisms that influence the species' survival. Our case study is Alaska sablefish (*Anoplopoma fimbria*), a highly valuable commercial groundfish species with extremely high movement rates and an early life history that begins offshore on the Alaskan continental slope. We hypothesize that the North Pacific Polar Front and the associated currents act as a conduit for transporting heat, salt, and nutrients from the western and central North Pacific to Alaska waters. This advection plays a key role in shaping the oceanographic climate of Alaskan waters and may influence the survival, and subsequent recruitment, of sablefish during their early life history.

Adult sablefish are distributed across the North Pacific from northern Mexico to the Bering Sea. Two populations exist within this range based on differences in growth rate, size at maturity, and tagging studies (McDevitt, 1990; Saunders et al., 1996;

Kimura et al., 1998). The northern or Alaska sablefish are assessed as a single population in Federal waters off Alaska due to their propensity for large-scale movements (Heifetz and Fujioka, 1991). Adult sablefish in Alaska are typically encountered between 200 and 1000 m along the continental slope, shelf gullies, and deep-sea canyons (Wolotira et al., 1993). In early spring, spawning and egg incubation take place at depth (> 300 m). Following hatch, larvae begin to feed and immediately swim to the surface (Mason et al., 1983). Larvae have been sampled in surface waters far from shore (160 km in southeast Alaska to 240 km in the Aleutians) and grow very quickly from 1.2 to 2 mm per day (Wing, 1997; Kendall and Matarese, 1987; Sigler et al., 2001). When larvae are small, diet consists mostly of copepod nauplii and eggs and as they grow diet shifts to mainly copepods and euphausiids (Yang and Nelson, 2000). There is no clear transition from larvae to young-of-the-year (YOY) or age-0 sablefish. However, large, pigmented pectoral fins are a diagnostic feature of larvae as they grow and both stages appear to be obligate surface dwellers (neustonic) as they drift to shore (Kendall and Matarese, 1987). Juvenile sablefish may also exhibit some thermal intolerance to very cold water (Sogard and Olla, 1998). Typically, by the end of the summer YOY sablefish reach nearshore bays, which serve as overwinter habitat until the following summer when juveniles begin movement to their adult habitat, arriving within three to four years (Rutecki and Varosi, 1997). The long-distance dispersal to nearshore habitat and rapid growth requirements throughout this pelagic early life history stage suggest that YOY sablefish may be more susceptible to the influence of large-scale oceanographic features.

The trans-Pacific Subarctic Gyre dominates the large-scale ocean circulation in Alaska waters. It is comprised of two cells termed the Western and Eastern Subarctic Gyres, WSAG and ESAG, respectively. The demarcation line between the WSAG and the ESAG is ill-defined yet believed to be approximated between the 170°E and 180°E meridians (Longhurst, 2007). Three main currents dominate the counter-clockwise flow of the ESAG, also termed the Alaska Gyre. Eastward transport along the southern limb is achieved by the North Pacific Current, which bifurcates into two broad currents as it reaches the American continent near Vancouver Island, Canada (Tomczak and Godfrey, 1994; Longhurst, 2007). From this point, the poleward limb of the Alaska gyre consists of the broad (300 km) Alaska Current which flows slowly (5–15 cm/s) along the northeastern and northern Gulf of Alaska (GOA) until its transformation near Prince William Sound (150°W) to the southwestward-flowing Alaskan Stream. This narrow (100 km), swift (100 cm/s) current continues westward along the Alaska Peninsula and Aleutian Islands until it gradually weakens west of 180°W (Weingartner, 2005). Along most of the Alaskan continental shelf, these main currents are paralleled by the inshore Alaska Coastal Current (ACC). This is a narrow, wind- and buoyancy-driven current that is mediated by downwelling-favorable winds and freshwater runoff (Weingartner et al., 2002, 2005). The circulation strength of the Subarctic Gyre and bifurcation latitude for the North Pacific Current are governed by the location and intensity of the Aleutian Low (Weingartner et al., 2009), which is the prominent geophysical feature associated with Alaska climate (Mundy and Olsson, 2005). Low-pressure storm systems making up the Aleutian Low are often generated in the western and central North Pacific and strengthen as they propagate eastward into Alaska waters (Weingartner, 2005).

The Subarctic–Subtropical Transition Zone separates the Subarctic Gyre from the warmer Subtropical Gyre and the northern and southern boundaries of this zone have the character of frontal regions (Roden, 1991). Ocean fronts are relatively narrow zones of enhanced horizontal gradients (e.g. temperature, salinity, etc.) that

separate broader zones of relatively uniform ocean properties (water masses) with different types of vertical structure (stratification) (e.g., Belkin and Cornillon, 2003). The northern frontal region of the Transition Zone is characterized by a well-defined subsurface temperature minimum, which can be used to delineate the Subarctic Zone proper (Favorite et al., 1976). This is the region of the Polar Front, which was reliably mapped by Belkin et al. (2002) using in situ hydrographic data from a network of repeat meridional oceanographic sections maintained largely by Japanese agencies since 1978. The Polar Front extends across the entire North Pacific along a slanted path, ascending from 42°N off Japan up to 57°N in the GOA, where the front falls back and joins the southwestward-flowing Alaskan Stream along the Alaska Peninsula (Fig. 1). This path implies that the Alaska climate system is on the receiving end of large-scale atmospheric and oceanic perturbations that occur in the western and central North Pacific. Therefore, we consider the trajectory of the Polar Front to serve as a convenient proxy for delineating the path of the major currents transporting oceanic properties (i.e. heat, salt, nutrients) into the Alaska Gyre.

As a first step in examining the influence of this advection on sablefish recruitment, we will evaluate the propagation of sea surface temperature (SST) anomalies along the Polar Front path. The SST characterizes the water temperature of the entire upper mixed layer (UML) of the ocean and provides information about the heat exchange at the air–sea interface (Martin, 2004; Lalli and Parsons, 1993). It is widely recognized that changes in SST influence biological events, either directly through individual processes such as metabolism and growth or indirectly through complex interactions that impact spawning time, food availability, and movement (Lalli and Parsons, 1993; Keyl and Wolff, 2008). We develop time series of SST anomalies along the Polar Front path and integrate these indices within an established sablefish population assessment model to investigate the relationship of SST with sablefish year-class strength. If anomalies suggest a poor environment through sub-optimal thermal conditions, we expect this to influence the survival of sablefish either directly or indirectly. For instance, advection of anomalous water masses (e.g., very cold or warm water) along the Polar Front path could alter the suitability of the offshore pelagic environment that

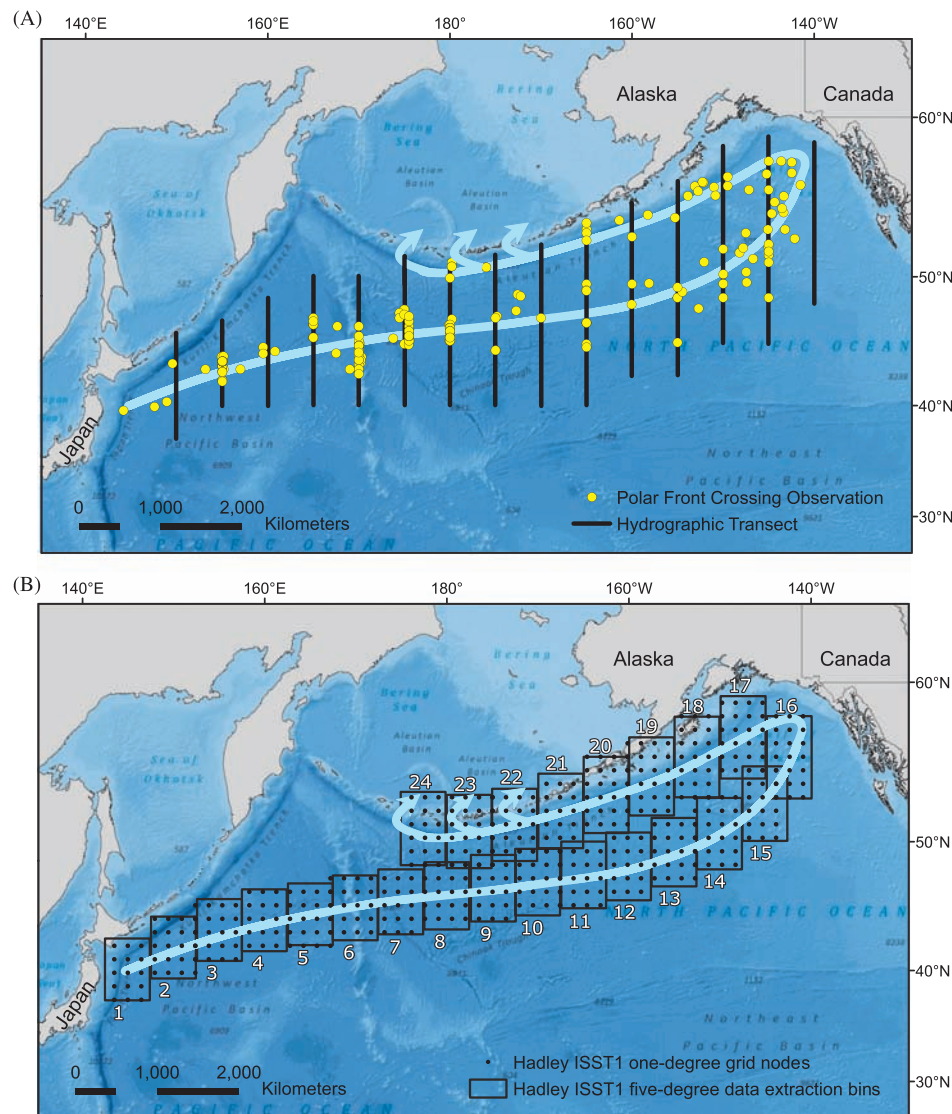


Fig. 1. Estimated mean path of the North Pacific Polar Front (blue line) in reference to (A, top panel) individual in situ Polar Front crossings (yellow dots) and hydrographic transects shown schematically (black lines) and (B, bottom panel) five-degree bins 1–24 (black rectangles) along the Polar Front with Hadley ISST1 one-degree grid nodes (black dots) shown schematically within the bins. Map was created using ArcGIS® 10 software by Environmental Systems Research Institute (ESRI), Redlands, CA. Background is Ocean Basemap and map projection is World Mercator. (For a color version of this figure, the reader is referred to the web version of this article.)

may subsequently influence sablefish development in Alaska waters. Our objectives are to (1) utilize the estimated location of the Polar Front path to generate time series of SST, (2) integrate biologically meaningful time series of SST into the sablefish stock assessment model, (3) evaluate competing models to determine the most important environmental covariate, and (4) determine the impact of these covariates for increasing precision of sablefish recruitment estimates.

2. Methods

SST measurements along the Polar Front may provide a good representation of the thermal events that impact the pelagic habitat experienced by neustonic sablefish larvae as they passively drift from the spawning region to the nearshore. SST measurements are derived from a variety of sources such as spaceborn infrared or microwave radiometers, moored or drifting buoys, research vessels, and ships of opportunity. SST gradients are associated with a variety of oceanographic features such as currents, eddies, and jets (Martin, 2004). Data available for estimating these properties along the Polar Front range from digital archives of reconstructed datasets (spatial and temporal interpolation of historical in situ observations) to collections of satellite imagery of remotely sensed radiances. We selected a long-term historical reconstructed dataset to describe the thermal conditions along the Polar Front path. The U.K. Met Office Hadley Centre ISST1 (HadISST1) global monthly SST dataset at one-degree spatial resolution is available from 1870 to present. This dataset combines cleaned, bias-adjusted historical in situ observations with satellite-derived measurements of SST using reduced space optimal interpolation methods to produce a gap-free dataset (Rayner et al., 2003). Data are freely available online through the Hadley Centre for Climate Prediction and Research.

2.1. Environmental time series development

We use the HadISST1 dataset to develop monthly time series of SST along the Polar Front path to investigate spatial and temporal patterns related to sablefish survival. Following Belkin et al. (2002), we updated the time series of in situ frontal boundary values with Japanese hydrographic data through 2009 and extended the Polar Front path along the Aleutians to 180°W (Fig. 1A). Instant locations of the Polar Front (yellow circles Fig. 1A) were identified from measurements taken along longitudinal hydrographic transects (red lines Fig. 1A) and a mean path was estimated between these locations for visualization purposes (blue line Fig. 1A). We then created data extraction bins at five by five degree resolution in order to extract time series to represent the SST data along the Polar Front path (Fig. 1B). Bin resolution was selected to capture the variability of the frontal crossings (yellow circles Fig. 1A) without overlapping adjacent bins along the path. Overall, 24 bins were created (black boxes Fig. 1B) and monthly SST data (black dots Fig. 1B) were extracted for each bin using the Mapping Toolbox in MATLAB (The Mathworks, Inc., Natick, Massachusetts). We then averaged the SST data by bin and month and calculated anomalies for each time series. These indices were integrated into the sablefish model to identify when and where the Polar Front was related to sablefish recruitment.

In addition to data extraction, we qualitatively analyzed the monthly SST data along the Polar Front path through Hovmöller diagrams. These are 2D surface maps that plot the time series of monthly SST anomalies for each of the 24 bins. The diagrams reveal propagation patterns of thermal anomalies along the Polar Front path through identification of propagation events and

estimation of propagation rates. Any event (i.e. warm or cold anomaly) that was advected eastward along the currents associated with the Polar Front would appear as a color streak slanted down from left to right as the event translated from the western to eastern Pacific. Additionally, an approximate propagation (advection) speed may be estimated from the distance the event traveled and the elapsed time. We summed the number of months that any particular anomaly lasted on the Hovmöller diagram and then traced the distance that anomaly traveled using the Mapping Toolbox. We present details on the spatial and temporal patterns of these propagation events and provide estimates of their propagation speed.

2.2. Sablefish model development

The current sablefish assessment model is a long-established integrated catch-at-age model which is updated annually and implemented with AD Model Builder (Fournier et al., 2012). Hereafter, we refer to this model as the base model. Many data sources are included in the base model. Fishery independent information (annual longline and trawl surveys) provide age/length compositions and survey abundance indices. Fishery dependent information includes foreign and domestic fishery catch, fishery age/length compositions, and catch-per-unit-effort indices. A few surveys have been conducted in the past to provide some information on the early life history stages of sablefish (Wing, 1997; Matarese et al., 2003). However, time series developed from this data are difficult to collate for use in a catch-at-age model due to the transient spatial and temporal nature of these surveys and the variety of gear types deployed. Following data assimilation, the analysis is completed in AD Model Builder which is C++ based software that uses automatic differentiation to optimize the fit of nonlinear statistical models (Fournier et al., 2012). The base model estimates model parameters simultaneously while allowing for missing data within a penalized maximum likelihood function and produces estimates of spawning stock biomass, recruitment, fishing mortality, and projections of future harvest scenarios. These results are used to determine the recommended acceptable biological catch (ABC) for the following year. The total allowable catch (TAC) quota set by management is almost always based on the recommended ABC from the base model. Future harvest projections are used to report on the status of the stock with respect to overfishing (Hanselman et al., 2010).

Recruitment estimates based on the current base model are extremely episodic and highly variable. A primary issue (as with many stocks) is determining whether this variability results from fluctuations in spawning stock size (i.e. egg production) or the environment (Quinn and Deriso, 1999). It has been observed that there is no apparent stock–recruitment relationship for many Alaskan groundfish (Dorn et al., 2010; Hanselman et al., 2009). Even when specifically comparing methods to forecast recruitment for pollock, spawning stock biomass was not a significant covariate in a single tested model (Lee et al., 2009). Alaska sablefish is no exception as several of the high recruitments occurred during periods of low observed spawning biomass; therefore, recruitment does not appear to be closely related to the level of spawning biomass. While catch of young sablefish prior to the survey may reduce their abundance before they are fully vulnerable to the survey gear, thus affecting recruitment, the observed large fluctuations in recruitment have occurred in spite of precautionary fishing levels. Because of these factors, recruitment is not modeled with a traditional stock–recruitment relationship. Rather recruitment is computed as mean recruitment (μ) with annual recruitment deviations (ε_t) (Hanselman et al., 2010). Recruits are estimated as two-year olds in the base model

because the traditional adult surveys do not select for these ages. Additionally, as mentioned previously, the disparate data sources on egg to juvenile life history ages do not as yet allow for reliable inclusion in the catch-at-age model.

Along with large fluctuations in recruitment, there is also high uncertainty in the earliest (1960–1970s) and most recent estimates where there is very little data besides catch available to the model (Hanselman et al., 2010). Since sablefish are long-lived, demographic data such as age compositions from the 1980s contain some information on recruitment. However, this is only inferred at advanced ages where age compositions tend to have more ageing error (Hanselman et al., 2012) and the cohorts have already been subject to many years of fishing mortality. The early recruitments are, therefore, uncertain compared to later recruitments estimated from age compositions that can track them through multiple early years of life where ageing error is low and fishing mortality has not been substantial. For the most recent estimates (2008–2010), there is limited size and age data available as these younger ages have not yet been selected by the survey or fishery (Hanselman et al., 2010). These earliest and most recent recruitment estimates are, therefore, excluded from the model when computing average recruitment for biological reference points and projecting future catch quotas.

Since the level of spawning biomass is not closely related to sablefish recruitment, we assume that only a minimum level of spawners is required to produce recruits and postulate that environmental conditions are the primary drivers of early life survival. Statistical catch-at-age models such as the sablefish assessment base model allow for straightforward inclusion of environmental time series to test these relationships and estimate annual recruitment deviations. A modified recruitment equation was incorporated into the base model following Maunders and Watters (2003) as

$$R_t = \mu \exp(\alpha + \beta X_t + \varepsilon_t) \quad (1)$$

where R_t is number of age-2 recruits in year t , μ is equal to mean recruitment, α is a scaling parameter, β is a free parameter that determines the magnitude of the index effect, X_t is an environmental index in year t , and ε_t are log-normally distributed recruitment deviations. The scaling parameter α is defined as

$$\alpha = \ln\left(\frac{n}{\sum \exp(\varepsilon_t + \beta X_t)}\right) \quad (2)$$

where n is the number of data points of the recruitment index. This scaling ensures that μ is equal to the mean over the whole time period by removing the bias from a non-normalized environmental time series. We assume the recruitment deviations are log-normally distributed and add a prior distribution to the associated negative log-likelihood. With this method, any index on any scale can be used. If that index has explanatory power for recruitment, then there will be a reduction in the magnitude of the estimated recruitment deviations as the environmental index accounts for the recruitment variability in each year. Hereafter, we refer to models with this recruitment modification as the environmental models.

2.3. Model integration and selection

Initially, spawning and early life history information from previous sablefish research was used to limit candidate SST time series to the appropriate lags so that results would be biologically meaningful. Only time series from the winter prior to the birth year through the fall of the birth year were included to capture ocean conditions from pre-spawning through settlement. Additionally, SST indices were restricted to the time period 1960–2010 as this was the period when abundance data becomes available in

the sablefish assessment model and recruitment was reasonably well-estimated (Hanselman et al., 2010). This subset of biologically meaningful SST indices was then integrated into the sablefish model following the multistage hypothesis testing procedure described in Deriso et al. (2008). First, candidate indices were screened by performing an asymptotic two-sided chi-square test on the recruitment negative log-likelihoods from the current sablefish base model to the environmental model. All candidate environmental models differ by a single parameter from the base model. Therefore, for a 5% significance level test ($P < 0.05$), two times the difference in the negative log-likelihood is compared to the critical value for the chi-square of 3.84 based on $\Pr(\chi^2_1 < 3.84) = 0.95$ (Deriso et al., 2008). Due to the asymptotic nature of this test and potential issues from unmodeled random effects, the chi-square statistic is expected to include a candidate by chance more often than 5%. Therefore, this test can be viewed as an objective method to limit the number of candidate models for subsequent significance testing. Values that exceeded the chi-square critical value were considered worth pursuing through a second more accurate and complicated randomization test (Deriso et al., 2008).

This randomization test essentially evaluates model fit in relation to what would occur by random chance and is likely a more reliable estimate of true significance because it overcomes the bias caused by unmodeled random effects (Deriso et al., 2008). For the candidate environmental covariates that were significant with the chi-square test, 200 randomly resampled datasets were generated for each covariate to ensure consistent distributional characteristics between the random dataset and the covariate (Deriso et al., 2008). Then the sablefish environmental model was run for each of the 200 randomly resampled datasets of each covariate that passed the chi-square test. The proportion of negative log-likelihoods obtained from randomizing the covariate index that fit the model better than the observed covariate index determined the probability that the index was significantly related to sablefish recruitment. The negative log-likelihood of the candidate index was compared to the five and one percentiles of the random dataset distribution for that candidate. Given the large number of candidate models and the resulting concerns for detecting spurious correlations (Type 1 error), we chose less than one percentile to be an important result. When models did not converge to a negative log-likelihood at least as good as the base model, we omitted it from the candidate models for comparison and adjusted the p -value accordingly (if the index is useless, β is inestimable but should be zero).

In conjunction with the multistage testing procedure, it is important to assess the reliability of an environment–recruit relationship with estimates of prediction error (Francis, 2006). One common technique is the procedure termed cross-validation which splits the data used in a statistical model, analyzes a subset (or training sample) to calculate new model parameters, and then validates that model with the remainder of the original dataset (Efron and Tibshirani, 1993). Unfortunately, the utility of this procedure is severely hindered in highly parameterized nonlinear models such as the current sablefish model. The confounding interaction between the recruitment deviations and the SST parameter does not allow for valid interpretation of the standard cross-validation techniques. However, given that there may be several equally plausible environmental models, the results of a cross-validation may confirm the consistency between these models and provide additional evidence for consideration. With this caveat in mind, we conducted a leave-one-out cross-validation using all the SST variables that passed the initial chi-square screening. In this type of cross-validation the environmental models are re-run iteratively by removing the i th year of the environmental covariate and then estimating recruitment for

the *i*th year given the resulting environmental relationship with that *i*th year data removed (Francis, 2006). The mean absolute error (MAE) is a more natural measure of average error than the root-mean squared error which is sensitive to variability of errors within the distribution (Willmott and Matsuura, 2005). The MAE weights all the errors equally and is less sensitive to large deviations (Kendall and Buckland, 1982). We computed the MAE across recruitment years for each index in three ways. First, we compared the predicted *i*th value with the value of that recruitment estimated in the full environmental model for that covariate. Second, we compared the predicted *i*th value with the value of that recruitment estimated in the base model. Third, we compared the predicted *i*th value with the value of that recruitment estimated in the base model for just the years 1960–1976 where the amount of data to estimate recruitment was more limited. We also computed these statistics for an environmental model that did not pass the chi-square test as a frame of reference for the magnitude of the MAE statistics.

Another potential assessment of the reliability of an environment–recruit relationship can be conducted through the use of a retrospective analysis which is commonly applied to catch-at-age models such as the current sablefish model (Hanselman et al., 2010). This type of analysis involves starting from some time period earlier in the model and successively adding data to test for consistency among parameter estimates as the new data are added (NRC, 1998). If the parameter estimates change substantially with new data or if there is persistent directional change in the estimates then there is likely retrospective bias in the estimates. In this study we were interested in the reliability of the most recent recruitments (last five years) predicted from the environmental model when less supportive data exist to inform the model on the magnitude of these recruitments. We examined this by running the environmental model up to a particular year and only using the survey and fishery data that was available up to that year (i.e. model run in 2005 only used data that was available up to 2005). We calculate the MAE from a comparison of the last five years of recruitment estimates from the environmental model in each retrospective run to the recruitments estimated in the 2010 base model, which we consider to be the closest to the true recruitments. As an example, recruitments for 2001–2005 from the 2005 retrospective model run were compared to the 2001–2005 recruitment estimates from the 2010 base model. This serves as a measure of the potential reduction in retrospective bias (i.e. a MAE value over 100% would mean that the environmental index increased retrospective bias). Due to the limited data in the most recent years, the base model does not estimate recruitment in the final or terminal year but rather uses an estimate of average recruitment (Hanselman et al., 2010). We, therefore, also computed the MAE of just the terminal year recruitment for each environmental index compared to the “true” 2010 base model estimates. This MAE was compared to the MAE computed by using average recruitment as the terminal year estimate compared to the 2010 base model estimates. The ratio of the two MAE estimates is a measure of the prediction error reduction versus assuming average recruitment in the terminal year (i.e. a value of 100% would mean that the index does no better than assuming average recruitment).

2.4. Impact evaluation

A final important aspect of the environment–recruit analysis is determining the impact or utility of incorporating the environmental covariate in an assessment model (Deriso et al., 2008). After careful assessment of the model selection procedures in the previous section, we select the best environmental model for this

analysis. One simple approach to determine impact is to compare the amount of unexplained variability from mean recruitment between the base and best environmental models. This is accomplished by comparing the MAE of the recruitment estimates between the two models.

It is also useful to consider the effect the environmental covariate has on other quantities within the model that may be of management interest (Deriso et al., 2008). For the sablefish model, we concentrate on the impact of the environmental covariate on female spawning biomass when there was limited or non-existent information from other data sources. As mentioned previously, this occurs during the early part of the time series (1960–1976), when age compositions containing these early year classes were subject to more ageing error because ageing error increases in older sablefish (Hanselman et al., 2012). There is also limited data during the most recent years (2009–2010), when there was only catch data and no information on the younger ages because they were not yet selected by the survey or fishery. We follow Deriso et al. (2008) for the impact analysis of the current recruitment estimates (1960–2010). First, the best environmental model is fit to the data. Then the model is rerun with the parameters from the best environmental model held fixed and the SST covariate set to zero. This simulates the condition of the environmental covariate set to the average of the time series since the environmental indices were initially calculated as anomalies. We term this model the impact model and it determines what the spawning biomass would have been without the influence of the fluctuating environmental covariate. We present the spawning biomass estimates for the base model and the impact model with respect to their relative difference from the best environmental model for the years 1960–2010.

The impact analysis of the most recent years was conducted slightly differently because the influence on projections of future spawning biomass should be taken into account. In the current assessment, the base model is projected twenty-five years into the future (2011–2025) in order to determine the status of this stock with respect to overfishing. In these future years, the majority of the model parameters are held fixed except for recruitment and fishing mortality. For recruitment, the base model does not estimate recruitment (age-2 fish) for the terminal year or for future projections. Instead, as mentioned previously, average recruitment is used for the terminal year and future recruitment deviations are randomly drawn from the distribution of past recruitments. This method provides an estimate of uncertainty, but yields average recruitment for future years (Hanselman et al., 2010). For developing the impact model of the recent years, we project forward fishing conditions similar to the current management of sablefish and use a fishing mortality rate that allows for harvesting the full catch quota. The inclusion of an environmental covariate in the model allows for the estimation of three additional recruitment years than what was estimated by the base model. This is because the terminal year was not estimated and the model estimates recruitment for age-2 fish (i.e. an SST value in 2010 provides data for recruits in 2012). In order to determine the impact of the three additional recruitment estimates (2010–2012), the future recruitment deviations are still drawn from historical recruitment deviations, but are allowed an additional deviation when there was an environmental data point. Future projections were estimated for the base model and the best environmental model. The impact model was then calculated following the same methods as described above for the years 1960–2010. We present comparisons between the spawning biomass estimates for the base and impact models in terms of the difference from the best environmental model for the years 2011–2025.

3. Results

The data extraction process developed monthly SST indices from each of the 24 bins to be used in the qualitative assessment of spatial and temporal patterns as well as integration into the sablefish model. In the following sections, we first present the results of the qualitative assessment identified in the Hovmöller diagrams and then provide details on the model integration, selection, and impact evaluation.

3.1. Spatial and temporal patterns

We observed many instances of spatial cohesiveness of warm and cold anomalies from west to east throughout the time period 1960–2010 (Fig. 2), essentially identifying propagating events. These observations can be expected since the along-front current ensures downstream translation of thermal signals that originate upstream. As an example of such an event, we highlight the large warm anomaly from bins 2 through 14 in 1966–67 (dashed black

line Fig. 2). Most events were not ocean-wide. Generally, we identified sporadic breakdown of the along-front connectivity between the western and eastern subarctic Pacific, often occurring in the central North Pacific. The exceptions were the regime shifts of 1976–77 and 1988–89 which manifested as trans-oceanic shifts of cold epochs. After the 1976–77 regime shift, the cold epoch ceased in the Northeast Pacific and simultaneously began in the Northwest Pacific persisting through 1988. The intensity of propagation events seemed to change following the 1988–89 shift with minor warm and cold events in the following two decades, generally restricted to either side of the Pacific. One exception may be the 1997 warm event which translated through the central to eastern North Pacific, and culminated in an exceptionally strong warm event in the GOA. More recently, two minor cold events in 1999 and 2007–08 and a series of strong warm events in 2002–2005 were observed in the GOA, likely of local origin. Finally, we estimated propagation speeds for many of these propagation events at 10–20 cm/s or approximately 9–17 km/day.

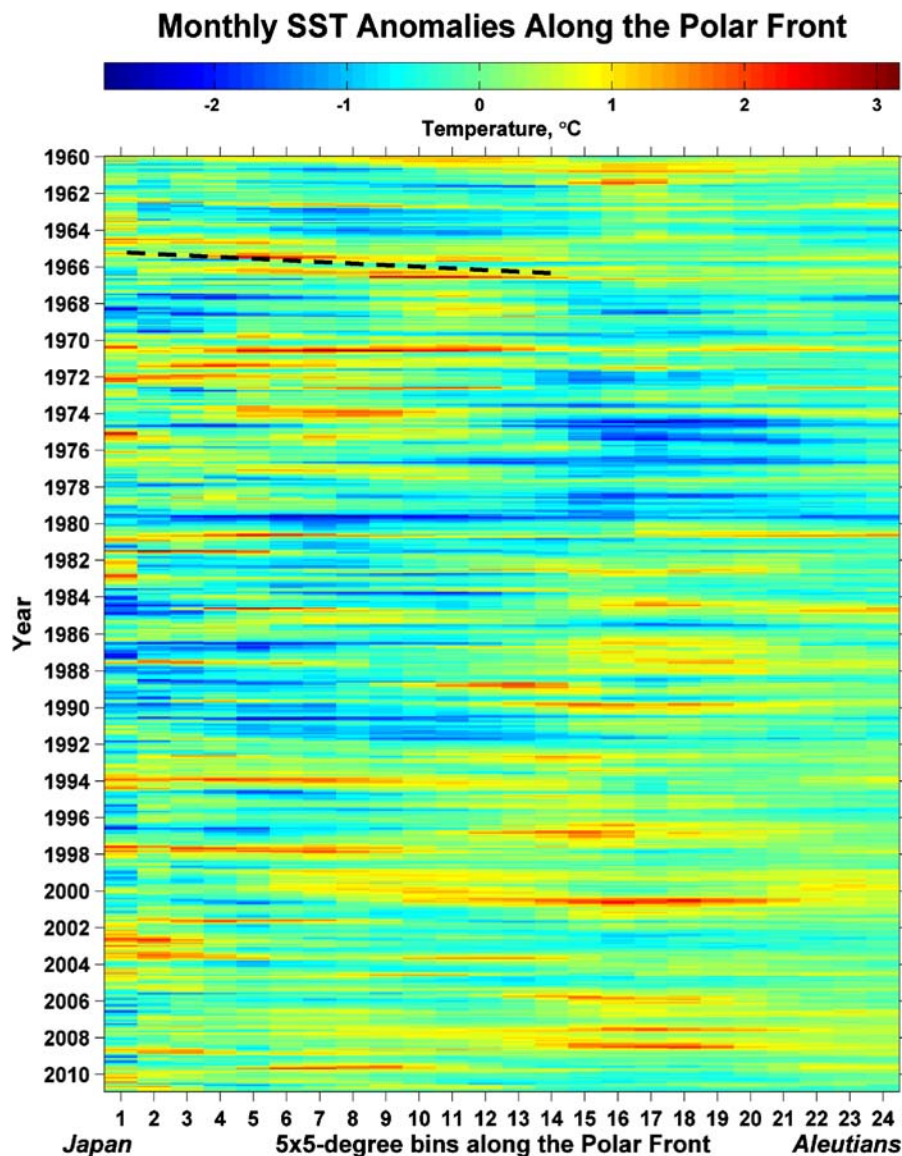


Fig. 2. Hovmöller diagram of monthly sea surface temperature (SST) anomalies, 1960–2010, along the Polar Front, from the Hadley ISST1 data. Horizontal axis shows bin numbers 1–24 along the Polar Front (see Fig. 1). Large warm anomaly of 1966–67 from bins 2 to 14 highlighted with dashed black line. (For a color version of this figure, the reader is referred to the web version of this article.)

Table 1

Significant indices using χ^2 and randomization tests in a two-stage process for Hadley ISST1 historical dataset. Parameters of model are provided along with test results. Base model and covariate that did not pass the selection tests are also provided.

Index name	Lag (year)	r-value	Beta	Beta CV	–lnL	χ^2	Randomization ($p < 0.01$)
Bin 13 December	–1	–0.51	–0.48	34%	11.15	7.55	Yes
Bin 12 December	–1	–0.46	–0.42	37%	11.86	6.11	Yes
Bin 14 December	–1	–0.44	–0.41	39%	11.91	6.01	Yes
Bin 13 November	–1	–0.43	–0.38	42%	11.35	7.14	Yes
Bin 12 November	–1	–0.41	–0.38	41%	12.21	5.42	Yes
Bin 11 December	–1	–0.41	–0.37	42%	12.40	5.04	No
Bin 14 November	–1	–0.38	–0.38	43%	12.22	5.39	No
Bin 11 November	–1	–0.37	–0.33	47%	12.68	4.47	No
Bin 5 October	–1	0.12	0.31	69%	14.60	0.63	Na
Base Model	NA	NA	0	NA	14.92	NA	Na

Table 2

Cross-validation and retrospective prediction error results for equally plausible models of Hadley ISST1 covariates that passed the χ^2 test. Results for a covariate that did not pass the test are also provided for reference.

Comparison	Cross-validation MAE			Retrospective error	
	To full	To base	To base (1960–1976)	Last 5 years compared to base	Terminal year to mean R
Bin 13 December	4.87	4.87	3.73	91%	53%
Bin 12 December	4.96	4.97	3.84	94%	50%
Bin 14 December	4.88	4.87	4.51	86%	51%
Bin 13 November	5.12	4.83	4.26	88%	25%
Bin 12 November	4.99	5.29	5.20	89%	36%
Bin 11 December	4.80	4.79	3.81	91%	50%
Bin 14 November	5.03	4.81	4.39	88%	28%
Bin 11 November	5.26	5.60	5.56	87%	40%
Bin 5 October	6.57	6.55	6.98	102%	73%

3.2. Model integration, selection, and evaluation

Restricting the potential SST indices to the period from pre-spawning to settlement resulted in 288 SST indices for further testing in the multistage procedure. The first stage chi-square test of the monthly SST time series yielded eight potential indices that exceeded the significance value of 3.84 (Table 1). These were wintertime SST (November and December) in bins 11–14 (central North Pacific) in the year before the sablefish were spawned. All relationships were negative and all estimates of the β parameter were of similar value and sign. Of the eight models that passed stage one, five were less than $p < 0.01$ in the randomization score for the stage 2 test (Table 1). These were wintertime SST in bins 12–14. The MAE estimates of prediction error from the leave-one-out cross validation were all very similar for the eight equally plausible models when comparing the predicted values to the full environmental model or the base model (Table 2). These values differed from the MAE for a model that did not pass the first stage testing (Bin 5 October). There was a slightly larger range of error estimated when comparing predicted values to the base for the early years (1960–1976). This likely reflects the higher uncertainty in the recruitment estimates for these years. Finally, the measures of reduction of retrospective bias for the eight equally plausible models were similar and about 10% better than the base model on average (Table 2). Estimates of MAE for the terminal year were all lower than the MAE of the base model in terms of prediction error from assuming average recruitment (Table 2). The model that did not pass the initial chi-square screening actually had an increased amount of retrospective bias and was not as good at reducing prediction error (Table 2).

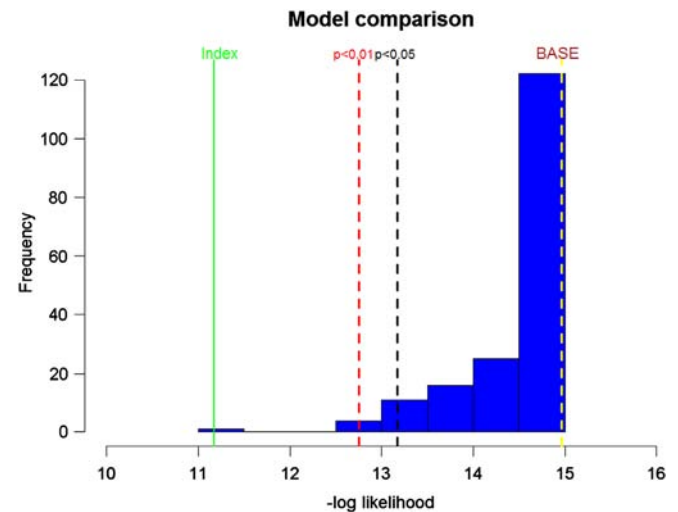


Fig. 3. Frequency of negative log-likelihood for the random time series (blue bars) compared to the negative log-likelihood when using the best candidate environmental time series (green "Index" line). Black and red dashed lines mark the 95th and 99th percentiles of the random distribution, respectively. Yellow dashed line is the negative log-likelihood of the base model. (For a color version of this figure, the reader is referred to the web version of this article.)

After evaluating these model selection tests for the equally plausible models, we chose the Bin 13 December model (Lag –1 year, $\beta = -0.48$, $p < 0.001$) as our best environmental model (Table 1). This model had the most negative correlation coefficient (r -value) with the recruitment index, the highest magnitude and lowest coefficient of variation (CV) for the β parameter, and the lowest negative log-likelihood over the other equally plausible models. The chi-square statistic for this model was the highest as can be expected since this statistic is based on the negative log-likelihood and the number of parameters does not differ between these models. For the randomization test, the negative log-likelihood obtained from this index (11.15) was substantially lower than the negative log-likelihoods obtained from all other randomizations of this index (Fig. 3). Additionally, a sensitivity screening of this model using 1000 randomization runs yielded approximately the same p -value (< 0.01). For the prediction error MAE, this model was consistent with the other equally plausible models in the leave-one-out procedure and had the lowest MAE estimate when comparing prediction error in the early recruitment years (1960–1976). This model was better than the base model in retrospective bias and prediction error of the terminal year, but was not the best model by these measures. Taken together, these model selection results imply that the Bin

13 December model is the best representative of the thermal environment in the central North Pacific that was related to sablefish recruitment. However, the other equally plausible models also had model selection results very similar to this index (albeit not the best) and the other covariates are in adjacent bins and months. These equally plausible indices covary and likely represent the same oceanographic process as the Bin 13 December index. Given these results, we consider the Bin 13 December covariate as the centroid of the oceanographic processes in the central North Pacific and we examine the impact of using SST covariates in the stock assessment model using this index.

The best environmental model (covariate=Bin 13 December) decreased estimated recruitment deviations by 17% from the base model. This suggests a moderate reduction in unexplained variability from mean recruitment with the best environmental model. A comparison of the recruitment estimates between the base model and the best environmental model highlights where this reduction has taken place (Fig. 4). When there were more age composition data and information on younger ages could be tracked through multiple cohorts (1977–2009), the recruitment estimates were nearly

identical between the two models (average CV base=36%, best=38%). However, when the data were subject to more ageing error (1960–1976) the estimates were more variable (Fig. 4A) and there was a slight decrease in uncertainty for the best environmental model (average CV base=96%, best=92%). This can be more easily seen when considering the percentage changes from the environmental model relative to the base model (Fig. 4B). For the impact analysis and the influence on female spawning biomass of the historical estimates (1960–2010), the effect was relatively slight for both the impact and base model but more pronounced in the early part of the time series for the base model (Fig. 5A). For future projections in the most recent years, the influence on projected future spawning biomass within the first five years was very low. Through the next several years, however, the influence of the SST index is much more prevalent (Fig. 5B). By helping predict the three additional recruitments (2010–2012), the SST index increased the spawning biomass when those age-classes reached maturity several years later in 2015. The influence declined after several years as the projection returned to average recruitment. This effect was more pronounced in the impact model because the SST estimates that

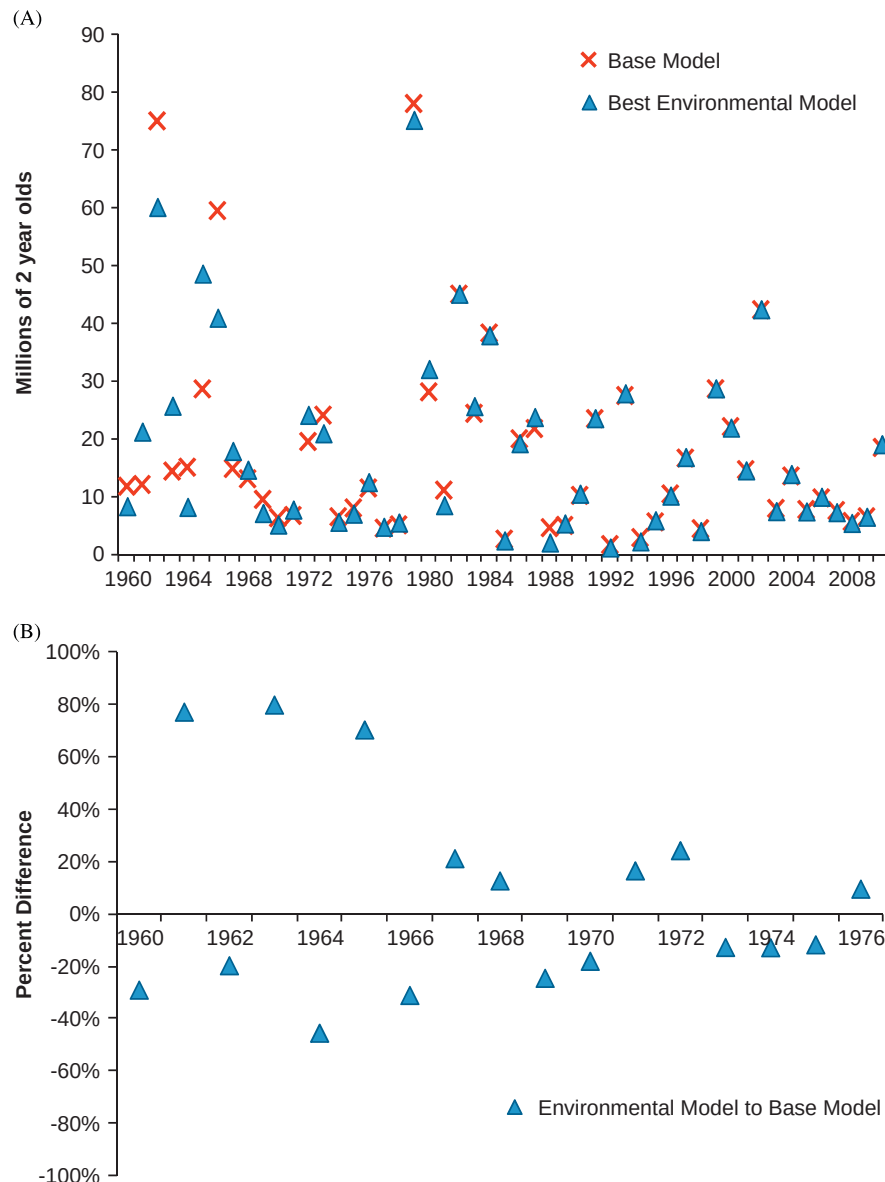


Fig. 4. Recruitment estimates from base and best environmental model for full time series from 1960 to 2012 (A) and percentage change of best environmental model relative to base model for the early part of time series with limited data from 1960 to 1976 (B).

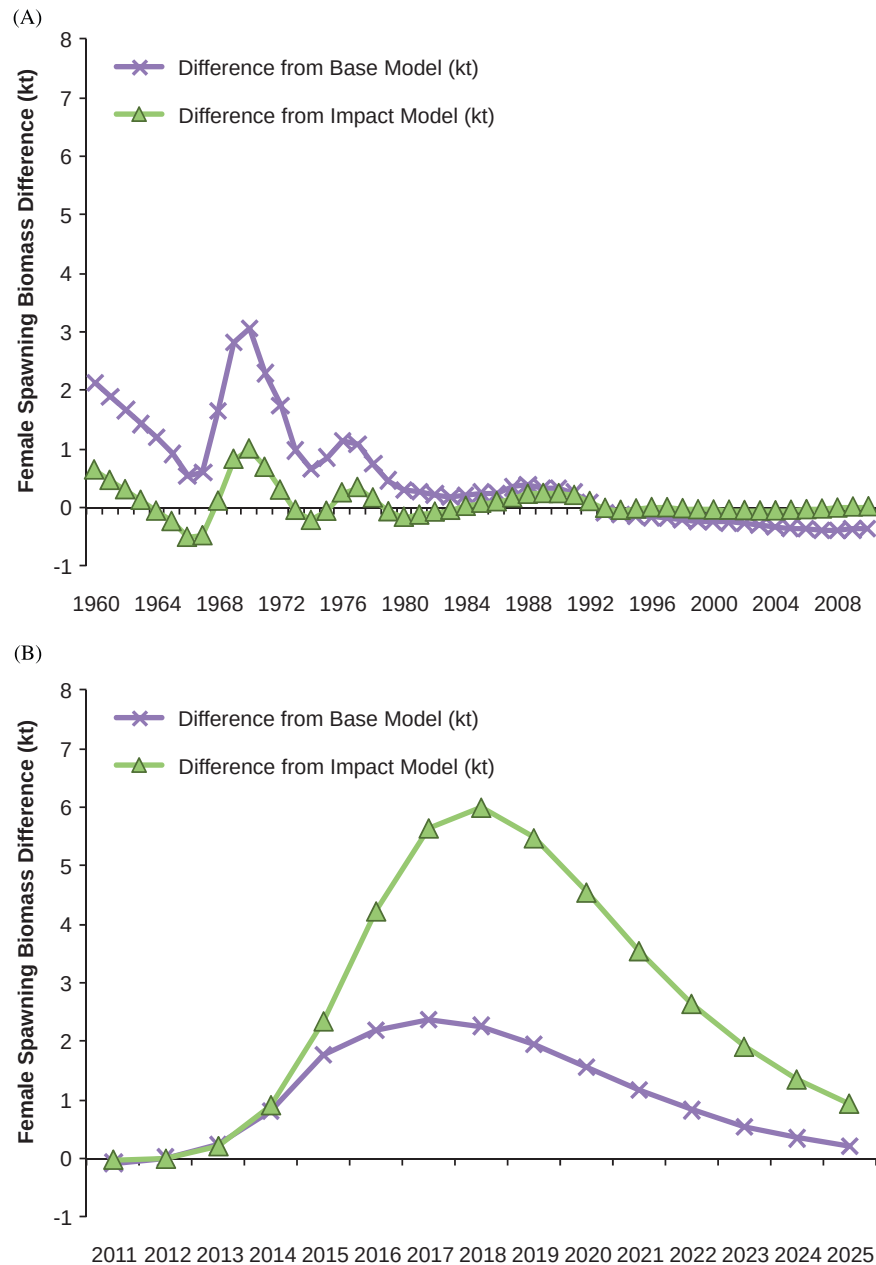


Fig. 5. Impact comparison of female spawning biomass estimates in terms of the difference between the best environmental model and the base model (purple line with x's) or impact model (green line with triangles) historically (A) and with future projections (B). (For a color version of this figure, the reader is referred to the web version of this article.)

indicated above-average recruitment in the environmental model were set to zero.

4. Discussion

Reducing uncertainty in the earliest and most recent recruitment estimates when there is little supportive demographic data from surveys or fisheries could increase the acceptability of these estimates for use in stock assessment. To accomplish this goal, we first developed indices that represent the spatial and temporal patterns of a proposed environmental feature influencing a given species' recruitment. We then performed a qualitative assessment of this oceanographic process using 2D surface maps. For our study, we inspected the propagation of thermal anomalies along

the Polar Front path through the North Pacific into Alaska waters. The overall nature of SST events seemed to be periods of spatial cohesiveness punctuated by sporadic breakdown typically in the central North Pacific with the exception of the trans-oceanic shifts of cold epochs at the onset of a regime. This sporadic pattern may manifest from meandering of the Polar Front due to atmospheric forcing. Following the visualization analysis, we integrated these indices within the sablefish population model and performed a series of model selection techniques to select the best environmental model. Wintertime indices of SST along the central North Pacific section of the Polar Front path were negatively related to sablefish recruitment. When combined with the qualitative observations, the identification of the location and season when SST along the Polar Front was related to sablefish recruitment allows for development of hypotheses on

the potential mechanisms that act to influence this species' survival.

The Aleutian Low is a dominant feature of Alaska climate and it affects the circulation pattern of the subarctic gyre. The position and strength of this pressure system generally result in two alternating warm and cool eras of winter atmospheric circulation in the eastern North Pacific (Hollowed and Wooster, 1992). A warm phase is typified by a more intense and eastward shifted Aleutian Low. Southerly winds associated with the eastern limb of the pressure system bring warm, moist air along the Alaska coast, while northerlies on the western limb bring cold Arctic air to the central North Pacific (Trenberth, 1990). As a result of this strengthening, the Alaska coastal region experiences warmer SST, higher precipitation, and increased downwelling (Weingartner, 2005). This may lead to an earlier onset of stratification which will decrease the depth of the mixed layer and subsequently provide a more intense and early spring bloom (Henson, 2007). Coastal discharge is also increased during the warm phase which will accelerate the flow of the Alaska Coastal Current (ACC) and enhance vertical mixing on the shelf. This could alter the onshore–offshore exchange of surface water resulting in increased onshore transport of nutrient-rich subsurface waters (Royer et al., 2001). Along the shelf break a more intense Aleutian Low may lead to strong changes in the mean velocity of the Alaskan Stream and in the associated mesoscale eddy field (Miller et al., 2005). Anticyclonic eddies (100–200 km) can significantly alter the cross-shelf exchange by trapping coastally derived nutrients in their interior and interacting with shelf-break currents. They may persist for several years, traveling hundreds of kilometers across the GOA and along the Aleutian Islands (Okkonen et al., 2003; Ladd et al., 2007). As eddies slowly decay, the entrained nutrients are injected into the surface layers increasing phytoplankton and zooplankton populations within the eddy and transporting fish larvae along the eddy path (Batten and Crawford, 2005; Crawford et al., 2007; Atwood et al., 2010). Finally, in the central North Pacific, the cold winds decrease SST while increasing turbulence and mixing. This will enhance upwelling bringing more nutrients to the upper layers and increasing the depth of the mixed layer. Both processes may act to support a higher primary and secondary production (Brodeur and Ware, 1992). The Alaska Current links the central North Pacific domain to the shelf/slope region of the GOA and likely transports the enhanced nutrients and plankton to this region (Ware and McFarlane, 1989).

The mean states of the Aleutian Low and subsequent changes in the biological community fluctuate on inter-annual and inter-decadal time scales (Hollowed et al., 2001). Large-scale climate indices such as the El Niño Southern Oscillation (ENSO), the Pacific Decadal Oscillation (PDO), and the North Pacific Gyre Oscillation (NPGO) have all been used to explain these fluctuations in the Alaska Gyre (Hollowed and Wooster, 1992; Mantua et al., 1997; Di Lorenzo et al., 2008). While these indices provide supportive evidence for overarching trends in the biological data, the identification of mechanisms influencing a particular stock remains unclear (Pyper et al., 2005). Additionally, the spatial scales of recruitment variation for some demersal fish stocks in the GOA ecosystem have been estimated at 1000–2000 km (Mueter et al., 2007), which suggests that regional indices may be more important to identifying environment–recruit relationships. In this study, we created relatively small bins along the Polar Front path to allow for the identification of the season and area most related to sablefish recruitment. Based on our model selection results, we considered the Bin 13 December covariate to be the best representation of the oceanographic processes in the central North Pacific. However the similarities in the model selection results of the other equally plausible models suggest

that these covariates represent the same process. Taken together then, the eight equally plausible covariates may represent a section of the large cold pool that forms during the winter in the central North Pacific as a result of an intensifying Aleutian Low (Mantua et al., 1997). The high wind stress of the descending limb may also shift the Polar Front to a more southerly position. The spatial extent of the four bins (11–14) representing the covarying SST indices is within the Alaska Gyre proper and consistent with that estimated by Mueter et al. (2007). Therefore, we suggest that this area likely represents the central North Pacific thermal environment resulting from the mean states of the Aleutian Low and serves as an indirect indicator of downstream effects on the suitability of the shelf/slope environment in which sablefish must survive.

The negative relationship described above implies that an intensifying Aleutian Low sets up conditions for increased survival of sablefish. Gargett (1997) proposed an optimum stability window to describe the compromise between opposing forces influencing water column stability to enhance phytoplankton growth. Here we take this concept one step further and suggest that in order to survive, sablefish larvae must also encounter optimum production domains as they travel through their pelagic early life history stages. We first follow Weingartner (2005) and divide the area sablefish must traverse into three basic domains based on characteristics of water mass structure and circulation. The outer shelf domain consists of the shelf-break and continental slope with the Alaska Current and Alaskan Stream driving the circulation in this region. The inner shelf domain is dominated by the ACC and stretches to the nearshore. These currents are continuous over the shelf/slope environment of the GOA and, therefore, advection strongly influences these domains (Weingartner, 2005). Finally the mid-shelf domain occurs between the inner and outer shelf domains and may be influenced by mesoscale eddies translating from their nearshore formation regions and current meanders from the outer shelf (Weingartner, 2005; Ladd et al., 2007). Both the dominant currents and eddy features of the three domains can efficiently translate downstream oceanic properties of heat, salt, nutrients, and plankton throughout Alaska waters (Weingartner, 2005). We next couple the physical conditions associated with an intensified Aleutian Low with the pelagic early life history ecological characteristics of sablefish to define a representative mechanism influencing sablefish survival for each domain.

We propose that a strong year class of sablefish relies upon the compounding effects of three separate mechanisms in what we term the Ocean Domain Dynamic Synergy (ODDS) conceptual model. Often strong year classes depend on a melding of several favorable environmental conditions during critical stages of development while poor year classes may occur during any of several unfavorable conditions (Hollowed and Wooster, 1992). The ODDS model for sablefish is initiated by a strengthening Aleutian Low. The ensuing first mechanism begins with the enhanced productivity in the central North Pacific cold pool (Mantua et al., 1997) from cold Arctic air in the descending limb of the Aleutian Low. This cold productive water is advected by the Alaska Current to the outer shelf domain of the GOA, arriving around spring (April–May). The timing of this arrival is based on an average propagation rate along the Polar Front path (~10–15 km/day). Sablefish larvae enter this outer shelf domain in spring with high consumption rates to meet their rapid growth requirements (Sigler et al., 2001). If there is a successful match between the timing of sablefish entering the outer shelf domain and the arrival of the cold pool productive waters, then sablefish may be able to utilize the productive water and increase survival through this domain. The second mechanism involves the increased anticyclonic eddy activity in the mid-shelf domain

due to the interaction between downwelling-favorable winds and the strengthening of the ACC and Alaskan Stream (Weingartner, 2005; Henson and Thomas, 2008). The increased circulation can influence the trajectory of eddies (Okkonen et al., 2003) and may result in trapping eddies as they translate along the shelf-break through the Alaskan Stream eddy corridor (Henson and Thomas, 2008). The prevailing theory for Alaska sablefish is that the majority of the spawners exist in the central and eastern GOA and these populations always produce a base level of recruitment. However, smaller populations of sablefish exist in the western GOA and Aleutian Islands (AI) and as larval sablefish from these populations enter the surface waters they may be entrained in coastally-derived eddies translating through this region. The development of large pectoral fins during this life stage (Kendall and Matarese, 1987) suggests that sablefish are designed to maintain their position in the water column and potentially capitalize on the productive surface waters within an eddy. This may provide a distinct advantage as the eddy contains higher levels of production which may increase the nutritional condition of entrained sablefish (Atwood et al., 2010). The increase in current speed and eddy activity may enhance the survival of the western GOA and AI populations as they may have a chance to grow quickly in the eddies and then be swept into protective nearshore bays or the more productive waters of the Bering Sea shelf. Finally, in the third mechanism the higher stratification along the coast from warmer SST and increased freshwater discharge may result in an early spring bloom that supports a large zooplankton biomass. The zooplankton production may be sustained through the summer by the onshore transport of nutrient-rich subsurface water (Royer et al., 2001; Henson 2007). The thermal intolerance of juvenile sablefish to very cold water may indicate the preference for warmer water to increase metabolism and enhance growth (Sogard and Olla, 1998). Also, as YOY sablefish enter the coastal waters, they are less passive and may begin to vertically migrate. The warmer surface waters and enhanced onshore transport of subsurface waters may allow YOY sablefish to increase growth by capitalizing on the larger zooplankton biomass throughout the water column and be transported quickly to their nearshore habitat.

The proposed ODDS conceptual model connecting sablefish survival and physical conditions resulting from a strong Aleutian Low are supported by several studies. The extremely large 1977–78 and 1980–81 year classes of sablefish occurred when biomass was near historic lows (Hanselman et al., 2010) suggesting sablefish was highly influenced by environmental forces acting during this time period. Hollowed and Wooster (1992) demonstrate that these successful recruitment years for sablefish coincide with strong year classes for many northeast Pacific groundfish stocks. Hare and Mantua (2000) attributed this high survival synchrony to a major regime shift in the North Pacific and associated it with a phase change to positive PDO conditions. Sigler et al. (2001) determined that YOY sablefish growth rate was higher in years when recruitment was above average and that above average recruitment was somewhat more likely with northerly winter currents and above average temperature which are associated with the coastal conditions during an intensified Aleutian Low. For the Canadian sablefish population, King et al. (2000) observed that years with above average recruitments were characterized by intense Aleutian Lows, more frequent southwesterly winds, and warmer coastal sea surface temperatures. Another study on U.S. West Coast sablefish by Schirripa and Colbert (2006) showed a significant positive relationship between sablefish recruitment, northward Ekman transport, eastward Ekman transport, and sea surface height. Finally, Sogard (2011) analyzed otolith increments of late larvae and early juvenile sablefish captured off the Oregon coast from 1993–2001 and

showed that growth rates were rapid and increased with warmer water temperatures. In this study, growth was positively related to a negative PDO index in February and reduced northerly transport in March. These findings were generally consistent with the large-scale oceanographic forcing of the California Current system. The opposite effects between the northern and southern sablefish populations are consistent with the inverse production regime forced by the Aleutian Low shifts, whereby higher production was associated with a positive PDO in the north and a negative PDO in the south (Mantua et al., 1997).

Investigation of other environmental indices may provide supportive evidence for the identified relationship with winter-time SST in the central North Pacific and allow for testing of the three proposed mechanisms in the ODDS model. Regional indices such as chlorophyll-*a*, sea surface height, and freshwater discharge may also be related to sablefish recruitment. Initially, we considered integrating high resolution satellite-derived chlorophyll-*a* indices within the sablefish population model to explore potential patterns with food availability. A simple correlation test between the chlorophyll-*a* indices and sablefish recruitment did indicate potential relationships along the extent of the Polar Front path. Unfortunately, the short-term nature of this data (1998 to present) caused inconsistencies in the sablefish model and was subsequently not useful for prediction. Environmental indices that encompass the potential cycles of variability in the recruitment estimates have a higher chance of correctly detecting environmental forcing (Haltuch and Punt, 2011). A relatively long-lived species such as sablefish (maximum age of 94 years) with a long catch history (1960s to present) may require a long-term index of environmental data for testing environment-recruit relationships within these more complex population models. For future studies, we recommend exploration of the shorter-term indices to support the ODDS conceptual model of the three proposed mechanisms influencing sablefish recruitment. Longer-term indices potentially developed through regional ocean models that adequately resolve the characteristics of the short-term indices may then be created. Relevant indices could be used in concert with the SST index to describe the optimum production regime and subsequently integrated and evaluated within the sablefish assessment model for further reduction of recruitment uncertainty.

Our intention for integrating an environmental covariate in the sablefish assessment model was to gain precision when there was limited information from other data sources in the model. The integration of the SST covariate within the base model resulted in a moderate reduction of recruitment uncertainty from the 2010 base model (17%) and provided information on the near-term future projection of sablefish spawning biomass. For the early recruitment estimates (1960–1976) there was only a slight increase in precision and impact to female spawning biomass. The three additional years of SST data increased future projections of recruitment than those projected by the base model. This affected the female spawning biomass several years later as those projected recruits became mature. Overfishing and harvest decisions in the North Pacific are based on the magnitude of spawning biomass relative to biological reference points. Therefore, the information gained from incorporating more reliable recruitment estimates into future projections may allow for increased efficiency in harvest decisions and rebuilding plans. In addition, more accurate medium term projections of stock trajectories may help guide capital expenditures and market pricing. Due to the high value of sablefish, even a small gain of ten metric tons would result in an annual ex-vessel value change of \$100,000 U.S. A highly correlated environmental index would provide some information as to the stock condition during the years prior to recruitment to the survey or fishery. Because of this, stakeholders

may be more amenable to utilizing the environmental information for projecting future catch quotas.

5. Summary and conclusions

This study provides a framework for developing indices of environmental features within the context of a species-specific stock assessment. The identified relationship between wintertime SST in the central North Pacific and sablefish recruitment provided insight for the development of the ODDS conceptual model which related biophysical conditions in three domains to sablefish early life history survival. Continued exploration of other environmental variables such as upwelling indices, freshwater discharge, and sea surface height anomalies within these three domains may provide supportive evidence for the identified mechanisms. Regional indices at the appropriate scales of covariation for a given species rather than large-scale indices such as the PDO may be more useful as consideration for transfer efficiencies through trophic levels and time lags may be more clearly interpreted to a mechanistic response. The impact of regime shifts on this species (and many other groundfish in Alaska) suggests that conditions that define a transition to a new regime may be extremely important to identify. Analysis of other environmental indices along the Polar Front path may provide early warning of such an event as seen with the SST diagrams. Propagation rates can be further verified for these events using the observed frontal crossings and an examination of the path meanders. Additional vetting and testing of incorporating environmental indices within the sablefish stock assessment will be required before potential implementation into a harvest strategy. However, the results presented here allowed for the development of hypotheses on three potential mechanisms influencing sablefish. This alone is useful information for fishery managers and provides a foundation for future studies investigating the ODDS model for sablefish recruitment. These methods are applicable to many other regions and species and the results may allow for more informed and efficient management decisions that stabilize catch and better achieve optimum yield.

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